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Question 1 [C01] [10 Points]

Design the WorldCupMatch class in such a way that the following code provides the expected output. [To win a match, a team has to score 3 or more goals.]

Driver Class	Output
<pre> public class WorldCupTester { public static void main(String[] args) { WorldCupMatch brazil = new WorldCupMatch(); WorldCupMatch norway = new WorldCupMatch(); brazil.setDetails("Brazil", "Marquinhos", true); norway.setDetails("Norway", "Odegaard", true); System.out.println("===== BRAZIL VS NORWAY ====="); brazil.scoreGoal(true); norway.scoreGoal(false); brazil.scoreGoal(false); System.out.println("====="); brazil.display(); System.out.println("-----"); norway.display(); System.out.println("====="); brazil.checkWinner(); norway.checkWinner(); System.out.println("====="); brazil.scoreGoal(true); brazil.checkWinner(); } } </pre>	<pre> ===== BRAZIL VS NORWAY ===== Brazil scored a goal! And got a Yellow. Norway scored a goal! Brazil scored a goal! ===== Team: Brazil Captain: Marquinhos Goals: 2 Yellow Cards: 1 Qualified: true ----- Team: Norway Captain: Odegaard Goals: 1 Yellow Cards: 0 Qualified: true ===== Brazil has not won yet. Norway has not won yet. ===== Brazil scored a goal! And got a Yellow. Brazil wins the match! Neymar scores the winning goal and celebrates with a Samba dance! </pre>

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Question 1 [C01] [10 Points]

Design the WorldCupMatch class in such a way so that the following code provides the expected output. [To win a match, a team has to score 4 or more goals.]

Driver Class	Output
<pre> public class WorldCupTester { public static void main(String[] args) { WorldCupMatch argentina = new WorldCupMatch(); WorldCupMatch egypt = new WorldCupMatch(); argentina.setDetails("Argentina", "Messi", true); egypt.setDetails("Egypt", "Salah", true); System.out.println("=== ARGENTINA VS EGYPT ==="); argentina.scoreGoal(false); egypt.scoreGoal(true); argentina.scoreGoal(false); argentina.scoreGoal(true); System.out.println("====="); argentina.display(); System.out.println("-----"); egypt.display(); System.out.println("====="); argentina.checkWinner(); egypt.checkWinner(); System.out.println("====="); argentina.scoreGoal(false); argentina.checkWinner(); } } </pre>	<pre> === ARGENTINA VS EGYPT === Argentina scored a goal! Egypt scored a goal! And got a Red. Argentina scored a goal! Argentina scored a goal! And got a Red. ===== Team: Argentina Captain: Messi Goals: 3 Red Cards: 1 Qualified: true ----- Team: Egypt Captain: Salah Goals: 1 Red Cards: 1 Qualified: true ===== Argentina has not won yet. Egypt has not won yet. ===== Argentina scored a goal! Argentina wins the match! Messi scores the winning goal and lifts the crowd to their feet! </pre>

SET A

```
public class WorldCupMatch {

    private String team;
    private String captain;
    private boolean qualified;
    private int goals;
    private int yellowCards;

    public void setDetails(String team, String captain, boolean qualified) {
        this.team = team;
        this.captain = captain;
        this.qualified = qualified;
        goals = 0;
        yellowCards = 0;
    }

    public void scoreGoal(boolean yellowCard) {
        goals++;

        if (yellowCard) {
            yellowCards++;
            System.out.println(team + " scored a goal! And got a Yellow.");
        } else {
            System.out.println(team + " scored a goal!");
        }
    }

    public void display() {
        System.out.println("Team: " + team);
        System.out.println("Captain: " + captain);
        System.out.println("Goals: " + goals);
        System.out.println("Yellow Cards: " + yellowCards);
        System.out.println("Qualified: " + qualified);
    }

    public void checkWinner() {
        if (goals >= 3) {
            System.out.println(team + " wins the match!");
            System.out.println("Neymar scores the winning goal and celebrates with a Samba dance!");
        } else {
            System.out.println(team + " has not won yet.");
        }
    }
}
```

```
}  
}  
}
```

SET B

```
public class WorldCupMatch {  
  
    private String team;  
    private String captain;  
    private boolean qualified;  
    private int goals;  
    private int redCards;  
  
    public void setDetails(String team, String captain, boolean qualified) {  
        this.team = team;  
        this.captain = captain;  
        this.qualified = qualified;  
        goals = 0;  
        redCards = 0;  
    }  
  
    public void scoreGoal(boolean redCard) {  
        goals++;  
  
        if (redCard) {  
            redCards++;  
            System.out.println(team + " scored a goal! And got a Red.");  
        } else {  
            System.out.println(team + " scored a goal!");  
        }  
    }  
  
    public void display() {  
        System.out.println("Team: " + team);  
        System.out.println("Captain: " + captain);  
        System.out.println("Goals: " + goals);  
        System.out.println("Red Cards: " + redCards);  
        System.out.println("Qualified: " + qualified);  
    }  
  
    public void checkWinner() {  
        if (goals >= 4) {  
            System.out.println(team + " wins the match!");  
            System.out.println(captain  
                + " scores the winning goal and lifts the crowd to their feet!");  
        }  
    }  
}
```

```

        } else {
            System.out.println(team + " has not won yet.");
        }
    }
}

```

RUBRICS:

Class implementation and correct instance variable	2 marks
setDetails()	-taking correct parameter (1 mark) - correctly updating (1 mark)
scoreGoal()	– correctly create method with proper parameter (0.5) – goal increment (0.5) – proper checking (2)
display()	– correctly create method with proper parameter (0.5) – correctly printing (0.5)
checkWinner()	- correctly create method with proper parameter (0.5) – proper checking (1.5)